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Product Catalog

Stainless Steel 17-4PH

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17-4PH is a widely-used martensitic precipitation-hardening stainless steel with a high concentration of chromium and nickel. It has many favorable properties, including excellent strength, good corrosion resistance, and excellent mechanical properties up to 600 °F (316 °C).

17-4PH also possesses good toughness when welded, and short low-temperature heat treatments reduce distortion and scaling. Additionally, 17-4PH is easy to machine and fabricate.

Due to the beneficial properties, 17-4PH finds use in many industries. Typical parts include mechanical seals, oil pump shafts, and pump shafts.

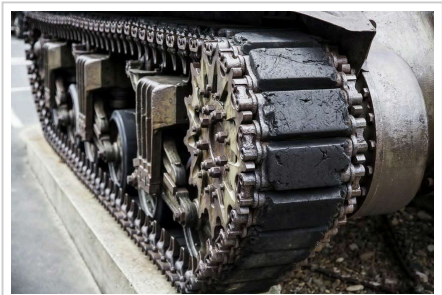
In addition, the alloy also finds use in the aerospace industry, petroleum industry, and chemical industry, food processing equipment, paper and pulp industry, nuclear waste processing and storage, and general engineering.

Tech Steel & Materials offers Stainless Steel 17-4PH in two sub-type specifications and multiple shapes/forms:

- AMS 5604 (Plate, Sheet, and Strip)
- AMS 5643 (Bar, Forging, Ring, Tubing, and Wire)

Stainless Steel 17-4PH Chemical Composition

Element	Min (%)	Max (%)
Carbon, C	—	0.07
Chromium, Cr	—	17
Copper, Cu	—	0.05
Iron, Fe	—	Balance
Manganese, Mn	—	1
Nickel, Ni	3.0	5.0
Nitrogen, N	—	0.01
Phosphorus, P	—	0.04
Sulfur, S	—	0.03
Tantalum + Niobium, Ta + Nb	0.15	0.45
Titanium, Ti	—	0.10
Zinc, Zn	—	0.01



Stainless Steel 17-4PH

Request a Quote

Name *

Company *

Email *

Phone *

Alloy spec *

17-4PH

Shape *

Size/Diameter *

Measure *

Inches

Length *

Measure *

Inches

Qty *

Comments

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clusively to the element mentioned, but that one predominates other elements that are used in smaller ies.

ication and Working Instructions

ss Steel 17-4PH reacts best to hot forming at 1742 to 2192 °F (950 to 1200 °C), with annealing at 76 °F and aging. The alloy can also be cold-formed in limited form, primarily to plates in the annealed on. However, stress corrosion resistance can be improved with re-aging.

more, 17-4PH can be cut using mechanical operations, such as abrasive waterjets, machining, iw, and shearing. However, plasma cutting isn't recommended on this alloy.

l has the best welding characteristics of competing stainless steel alloys. Moreover, it can be welded ll standard techniques, including gas tungsten arc, gas metal arc, shielded metal arc, and plasma arc. To desired mechanical properties, pre-weld and post-weld heat treatments are recommended. For the best we recommend the AMS 5643 wire.

ical Properties of Stainless Steel 17-4PH

erty	Imperial	Metric
g Range	2560-2625 °F	1404-1440 °C
fic Heat	0.11 Btu/lb.-°F	460 Joules/gg-K
fic Density	0.282 lb/in3	7.8 g/cm3
ical Resistivity	38.6 μΩ in	98 μΩ cm
ρ Coefficient of Thermal Expansion (70 °F to 200 °F) (21 °C to)	6 [in/in/°F·106]	10.8 [μm/m·°C]
ρ Coefficient of Thermal Expansion (70 °F to 400 °F) (21 °C to)	6 [in/in/°F·106]	10.8 [μm/m·°C]
ρ Coefficient of Thermal Expansion (70 °F to 600 °F) (27 °C to)	6.2 [in/in/°F·106]	11.2 [μm/m·°C]
ρ Coefficient of Thermal Expansion (70 °F to 800 °F) (21 °C to)	6.3 [in/in/°F·106]	11.2 [μm/m·°C]
thermal Conductivity (@ 300 °F) (@ 149 °C)	124 Btu/(hr/ft²/in/°F)	17.9 [W/m-K]
thermal Conductivity (@ 500 °F) (@ 260 °C)	135 Btu/(hr/ft²/in/°F)	19.5 [W/m-K]
thermal Conductivity (@ 860 °F) (@ 460 °C)	156 Btu/(hr/ft²/in/°F)	22.5 [W/m-K]
thermal Conductivity (@ 900 °F) (@ 482 °C)	157 Btu/(hr/ft²/in/°F)	22.6 [W/m-K]
on's Ratio (H900 Condition)	0.272	0.272
ulus of Elasticity (H900 Condition)	28 x 10 ⁶ ksi	197 x 10 ³ MPa
ulus of Rigidity in Torsion	9.68 x 10 ³ ksi	67 x 10 ³ MPa

ainless Steel 17-4PH Mechanical Properties

erty	Imperial	Metric
ate Tensile Strength	23,300 psi	160 MPa
Strength (0.02%)	16,680 psi	115 MPa
ation (% in 2")	5%	5%
vell Hardness C	35	35

: Treatment

ss Steel 17-4PH is available in the solution annealed condition at 1900 °F (1038 °C) and then air-cooled °F (32 °C).

Metal Weight Calculator

Make use of our online metal weight calculator calculate the weight of Stainless Steel 17-4PH

Material

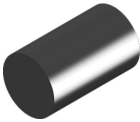
Stainless Steel

Alloy

17-4PH

Shape

Bar



Diameter

Inches

Length

Inches

Quantity

1

Weight

0 Lbs

CALCULATE

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Metal Weight Calculator - [Disclaimer](#)

er, additional heat treatments can produce various hardness and toughness levels. The most common
nts are listed in the table below.

tion	Temperature [± 15 °F (± 8,4 °C)]	Cooling Method and Duration
	900 °F (482 °C)	Air cooling for 1 hour
	925 °F (496 °C)	Air cooling for 4 hours
5	1025 °F (551 °C)	Air cooling for 4 hours
5	1075 °F (580 °C)	Air cooling for 4 hours
0	1100 °F (593 °C)	Air cooling for 4 hours
0	1150 °F (621 °C)	Air cooling for 4 hours
0 + 1150	1150 °F (621 °C) followed by 1150 °F (621 °C)	Air cooling for 4 hours followed by Air cooling for 4 hours
0-M	1400 °F (760 °C) followed by 1150 °F (621 °C)	Air cooling for 2 hours followed by Air cooling for 4 hours

AMS Number:

Number	Alloy	Type	UNS	Cross Ref. Spec	Misc./Shape
5604	17-4PH	Stainless Steel	S17400		
5604 Plate	17-4PH	Stainless Steel	S17400	-	Plate 
5604 Sheet	17-4PH	Stainless Steel	S17400	-	Sheet 
5604 Strip	17-4PH	Stainless Steel	S17400	-	Strip 
5643	17-4PH	Stainless Steel	S17400		
5643 Bar	17-4PH	Stainless Steel	S17400	-	Bar 
5643 Forging	17-4PH	Stainless Steel	S17400	-	Forging 
5643 Ring	17-4PH	Stainless Steel	S17400	-	Ring 
5643 Tubing	17-4PH	Stainless Steel	S17400	-	Tubing 
5643 Wire	17-4PH	Stainless Steel	S17400	-	Wire 



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| | Tungsten |

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